

say, a transistor and then solder on a new transistor immediately, saving you time in the process.

You can de-solder and re-solder several components simultaneously—quickly de-solder a problematic electrical part from the circuit board and re-solder the area again in a few minutes. In addition, hot-air stations are typically non-contact, enabling you to prevent damage to other parts of the circuit as you work.

However, one drawback of using hot-air rework stations is that these are much more expensive than de-soldering stations. Rework stations are also considerably bulky and difficult to carry around, so you will need to bring your work to the station itself, which can be difficult at times.

De-soldering stations

De-soldering stations are available with a variety of features; some include vacuum pumps that enable you to precisely de-solder whatever you want. Because SMT parts are usually small, de-soldering stations are also much easier to clean and maintain than hot-air rework stations.

De-soldering stations come packed with a lot of power, yet have fewer controls and nozzles than hot-air rework stations. But the usual drawback is that there is a greater risk of damage from voltage leakage when using de-soldering stations.

The most significant benefit in choosing a de-soldering station is probably its lower price tag; you can save upwards of a few hundred rupees by opting for a de-soldering station over a hot-air rework station.

Further, de-soldering stations are easier to move than rework stations. Although, you can get your job done with just a soldering iron and de-soldering station, a hot-air rework station may certainly be better for you if you are going to use most of its special features.

Bench-top de-soldering or rework stations

Generally, bench-top stations have the most accurate temperature control, best thermal recovery and other features like programmable timing and hot air combined with suction. If you plan to work with surface-mount ICs on a regular basis, rework units are ideal. These have a set of separate nozzles for just about every IC package. The higher-end units can be programmed to pre-heat for a specific period before applying heat and then heated for a number of seconds, and some even have a vacuum nozzle to pick up freed compo-

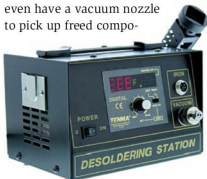


Fig. 2: De-soldering station

nents from the PCB using suction. But since these are costly, hobbyists are usually able to do without these.

Moreover, a component with one or two connections to the PCB can usually be removed easily by heating its joints with a soldering iron. Solder-filled hole can be cleaned out with a de-soldering pump or with a pointed object made of a material that the solder does not make wet, such as stainless steel, graphite or wood. If a multi-pin component does not have to be salvaged, it is usually possible to cut its pins and remove the residual ends by pulling these one by one.

Components with more pins cannot be extracted intact in the way described above, unless the wire leads are long and flexible enough to be pulled out. For a component such as a dual-in line package (DIP), the pins are too short to pull,

and solder melted on one joint will solidify before it can be melted on another pin.

Basics tips to observe while buying

Different products are being produced by all reputed manufacturers to meet user requirements at different levels of work situations. The products range from economic, entry-level stations with basic features, such as analogue thermostat based temperature control, to high-end expensive and most sophisticated units having precision digital temperature control and most sought-after component removal attachments. Since the station you are going to purchase would become an integral part of your toolkit, it is important to consider some points before making a purchase. The following tips can be useful:

Reliable brands. It is important to choose a well-reputed high-quality manufacturer's product. It is probably one of the most important decisions to make when you decide to make a purchase. A few of the premier imported brands in the solder industry include JBC Tools, Weller and Kester, Hakoo and Sumitron. Picking quality brands where it matters gives you the general incentives of superior customer support, trusted materials, manufacturing, longevity of solder tools and versatility to interchange solder irons and other supplies within your station seamlessly.

Control units. If you are working on a job with a high temperature requirement for soldering, you will need a comprehensive temperature control unit to help keep the temperature well under control. With a choice of proper control unit, you can accurately monitor the heat of the soldering iron to ensure that you keep the temperature at the required level, so that you can get the best results.

Always choose a model that has